

ARTIFICIAL INTELLIGENCE (AI) ASSIGNMENT #02

Section 1: Simple Linear Regression

- QUESTION # 01 -

Given:

$$X = [1, 2, 3, 4, 5] \quad Y = [2, 3, 5, 7, 11]$$

- (a) Compute Slope (β_1) and intercept (β_0)

$$n = 5$$

$$\sum X = 1 + 2 + 3 + 4 + 5 = 15$$

$$\sum Y = 2 + 3 + 5 + 7 + 11 = 28$$

$$\sum XY = (1 \cdot 2) + (2 \cdot 3) + (3 \cdot 5) + (4 \cdot 7) + (5 \cdot 11) = 2 + 6 + 15 + 28 + 55 = 106$$

$$\sum X^2 = 1 + 4 + 9 + 16 + 25 = 55$$

Slope β_1

$$\beta_1 = \frac{n \sum XY - \sum X \sum Y}{n \sum X^2 - (\sum X)^2} = \frac{[5 \times 106 - 15 \times 28]}{[5 \times 55 - 15^2]} = \frac{110}{50} = 2.2$$

Intercept β_0

$$\beta_0 = \frac{\sum Y - \beta_1 \sum X}{n} = \frac{28 - 2.2 \times 15}{5} = \frac{28 - 33}{5} = \frac{-5}{5} = -1$$

$$\text{Regression Equation: } \hat{Y} = -1 + 2.2X$$

- (b) Predicted Y when $X = 6$

$$\hat{Y} = -1 + 2.2 \times 6$$

$$\hat{Y} = -1 + 13.2 = 12.2$$

- (c) Compute R^2

$$\sum Y^2 = 4 + 9 + 25 + 49 + 121 = 208$$

$$\bar{Y} = \frac{\sum Y}{n} = \frac{28}{5} = 5.6$$

$$\begin{aligned} \text{SS}_{\text{Total}} &= \sum (Y_i - \bar{Y})^2 = \sum Y^2 - n \bar{Y}^2 = 208 - 5(5.6)^2 \\ &= 208 - 156.8 \\ &= 51.2 \end{aligned}$$

(2)

Compute Predicted Values

$$\hat{y} = [-1 + 2 \cdot 2 \times 1, -1 + 2 \cdot 2 \times 2, -1 + 2 \cdot 2 \times 3, -1 + 2 \cdot 2 \times 4, -1 + 2 \cdot 2 \times 5]$$

$$\hat{y} = [1.2, 3.4, 5.6, 7.8, 10]$$

$$\begin{aligned} \text{SS}_{\text{res}} &= \sum (y_i - \hat{y}_i)^2 = (2 - 1.2)^2 + (3 - 3.4)^2 + (5 - 5.6)^2 + (7 - 7.8)^2 + (11 - 10)^2 \\ &= 0.64 + 0.16 + 0.36 + 0.64 + 1 \\ &= 2.8 \end{aligned}$$

$$R^2 = 1 - (\text{SS}_{\text{res}} / \text{SS}_{\text{total}})$$

$$= 1 - \frac{2.8}{51.2} \approx 1 - 0.0547 \approx 0.9453$$

- QUESTION #02 -

Given:

$$y_{\text{actual}} = [3, 6, 8, 11]$$

$$y_{\text{predicted}} = [2.8, 5.5, 8.2, 11.1]$$

Compute MSE (Mean Squared Error)

$$\text{MSE} = (1/n) \sum (y_{\text{actual}} - y_{\text{predicted}})^2$$

$$= \frac{[3 - 2.8]^2 + [6 - 5.5]^2 + [8 - 8.2]^2 + [11 - 11.1]^2}{4}$$

$$= \frac{0.04 + 0.25 + 0.04 + 0.01}{4} = \frac{0.34}{4} = 0.085$$

Compute RMSE

$$\text{RMSE} = \sqrt{0.085} \approx 0.2915$$

- For $x=1$: 0.2
- For $x=2$: 0.5
- For $x=3$: -0.2
- For $x=4$: -0.1

Errors less than 1 in magnitude, small MSE value 0.085 resulted.

- QUESTION # 03 -

Reason:

Model B performs better because it has lower MSE 2.3 as compared to Model A 4.5.

MSE measures the average squared difference between predicted and actual values, so a lower MSE indicates better model performance with smaller prediction errors.

Section 2: Multiple Linear Regression

- QUESTION # 04 -

Given:

$$x_1 = [1, 2, 3, 4] \quad , \quad x_2 = [2, 3, 4, 5] \quad , \quad y = [3, 6, 9, 12]$$

- (a) Regression Equation

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$$

- (b) Compute β using normal equation

Construct matrix X and Y

$$X = [(1, 1, 2), (1, 2, 3), (1, 3, 4), (1, 4, 5)]$$

$$Y = [(3), (6), (9), (12)]$$

- compute $X^T X$:-

$$= [(1, 1, 1, 1), [(3), [(1, 30), [1, 2, 3,$$

$$\beta = (X^T X)^{-1} X^T Y$$

Construction of Matrices X and vector Y

$$X = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \\ 1 & 4 & 5 \end{bmatrix} \quad , \quad Y = \begin{bmatrix} 3 \\ 6 \\ 9 \\ 12 \end{bmatrix}$$

(4)

• First column = 1 (for intercept β_0)

• Second column = x_1

• Third column = x_2

2. Compute $X^T X$

$$X^T X = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \\ 1 & 4 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 10 & 14 \\ 10 & 30 & 42 \\ 14 & 42 & 54 \end{bmatrix}$$

3. Compute $X^T Y$

$$X^T Y = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 6 \\ 9 \\ 12 \end{bmatrix}$$

$$X^T Y = \begin{bmatrix} 30 \\ 90 \\ 126 \end{bmatrix}$$

4. Solve $X^T Y = (X^T X) \beta = X^T Y$

$$\begin{bmatrix} 4 & 10 & 14 \\ 10 & 30 & 42 \\ 14 & 42 & 54 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 90 \\ 126 \end{bmatrix}$$

• $y = 3x_1$, $(1, 3, 6, 9, 12)$ are exactly $3 \times (1, 2, 3, 4)$

• $x_2 = x_1 + 1$

Relationship

$$y = 0 \cdot \beta_0 + 3 \cdot x_1 + 0 \cdot x_2$$

(5)

5. Solving the system, using row reduction

$$\beta = (X^T X)^{-1} X^T Y$$

sol:

$$\beta_0 = 0, \quad \beta_1 = 3, \quad \beta_2 = 0$$

$$Y = 0 + 3X_1 + 0X_2 = 3X_1$$

which matches given $Y = [3, 6, 9, 12]$

• Predict Y for $X_1 = 5, X_2 = 6$

using regression Equation

$$\hat{Y} = 0 + 3(5) + 0(6)$$

$$\hat{Y} = 15$$

• $Y = X_1 + X_2$

but

• $Y = 3X_1$

- X is $3X_1$, where X_2 is irrelevant

- β is $[0, 3, 0]^T$

ANSWERS:

(a) $Y = 0 + 3X_1 + 0X_2$

(b) $\beta = \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix}$

(c) $\hat{Y} = 3 \times 5 = 15$

- QUESTION # 05 -

Reasoning:

No, a higher R^2 doesn't always mean a better model.
Adding variables can overfit the data. Use adjusted R^2 or cross validation.

- QUESTION # 06 -

Answer:

- Effect : Multicollinearity inflates variance of coefficients ^{estimates}
- Detection : High Variance Inflation Factor ($VIF > 5$)
- Solution : Remove correlated predictors or use Ridge Regression.

Section 3: Polynomial Regression

-QUESTION # 07-

Given :

$$x = [1, 2, 3, 4] \quad , \quad y = [2, 5, 10, 17]$$

(a) Transformed Equation (degree = 2)

$$X_{poly} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \end{bmatrix} \quad , \quad y = \begin{bmatrix} 2 \\ 5 \\ 10 \\ 17 \end{bmatrix}$$

(b) Solve for β

$$\beta = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$y = 1 + x^2$$

(c) Predict y for $x=5$:

$$\hat{y} = 1 + 0 \times 5 + 1 \times 25$$

$$\hat{y} = 26$$

-QUESTION # 08-

Reasoning: Model B (degree = 4) overfits the training model, leading to higher test error than Model A (degree = 2)

-QUESTION # 09-

Reasoning: Overfitting occurs when the model fits noise.
Regularization penalizes w -coefficients to improve generalization.

Section 4: Logistic Regression

- QUESTION #10 -

Given:

$$z = -2 + 0.8x$$

$$x = 3$$

Compute Probability:

$$p = \frac{1}{1 + e^{-(z)}} = \frac{1}{1 + e^{-(-2 + 0.8 \times 3)}} = \frac{1}{1 + e^{-0.4}} \approx 0.5987$$

- QUESTION #11 -

Given:

$$y_{\text{actual}} = [1, 0, 1, 1]$$

$$y_{\text{predicted}} = [0.9, 0.2, 0.8, 0.7]$$

Log Loss:

$$-\frac{1}{4} [\ln(0.9) + \ln(0.8) + \ln(0.7) + \ln(0.8)] \approx 0.2271$$

- QUESTION #12 -

Gradient of Log Loss:

$$\nabla_{\beta} L = X^T (\sigma(z) - y)$$

- QUESTION #13 -

Gradient Descent Update:

• Compute $\sigma(z)$ for each row

• Update β

$$\beta_{\text{new}} = \begin{bmatrix} -1 \\ 0.5 \end{bmatrix} - 0.01 \begin{bmatrix} -0.3288 \\ -0.3023 \end{bmatrix} \approx \begin{bmatrix} -0.9967 \\ 0.5030 \end{bmatrix}$$
